PARTICULAR SPECIFICATIONS P4D

Appendix P4D – THP Feed Parameters

CONTENTS

NOTE: THIS DOCUMENT SUPERCEDES THE ORIGINAL VERSION OF THE APPENDIX P4D – THP FEED PARAMETERS AT THE CLOSE OF TENDER. THE CHANGES HAS BEEN HIGHLIGHTED IN YELLOW.

P4D.1. TWRP BIOSOLIDS FEED PARAMETERS AND THP

(a) Figure P4D-1 shows the overall treatment schematic for TWRP biosolids plant and the various sources of feed to anaerobic digester. The digestion pre-treatment THP plant shall play an important role to treat all the WAS flows and up to 30% of thickened primary sludge from Domestic Liquids Modules (DLM). The total predicted WAS production from the DLM and Industrial Liquids Modules (ILM) is expected to contribute approximately 30% of the total (PS + WAS) sludge production for TWRP (as per Error! Reference source not found.).

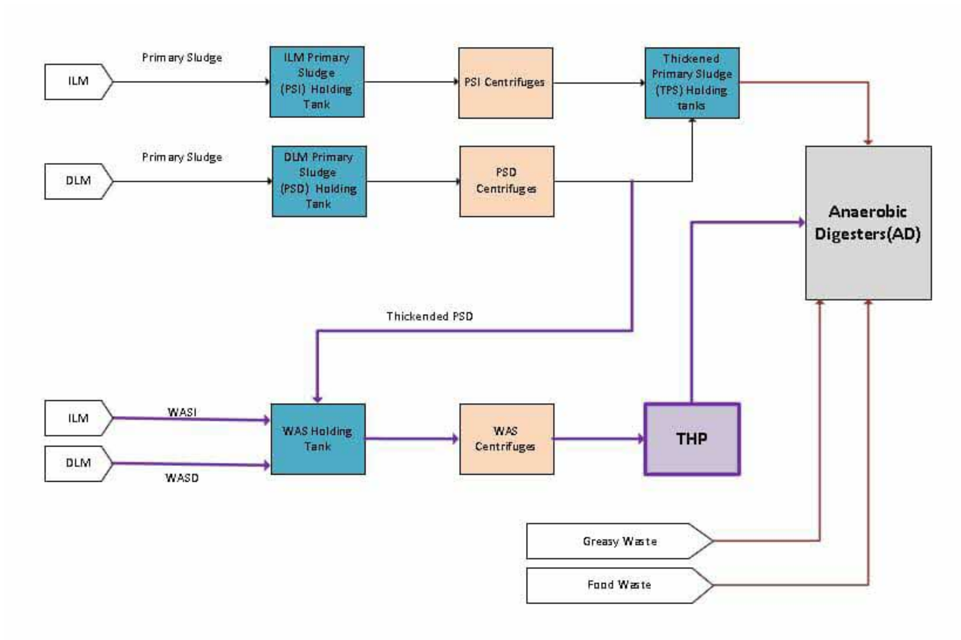

Figure P4D-1: Biosolids Treatment

(b) The design basis for the Biosolids treatment facility (Phase 1) is shown in Table P4D-1.

Table P4D- 1 Biosolids Treatment Phase 1 Design Basis

PARAMETER	ANNUAL AVERAGE		MAX MONTH	
	TOTAL SOLIDS (tDS/d)	DS (%)	TOTAL SOLIDS (tDS/d)	DS (%)
Domestic PS	166	0.85	214	0.85
Domestic WAS	73.4	0.67	85	0.67
Industrial PS	50	1.66	65	1.66
Industrial WAS	48	0.75	77	0.75
Greasy Waste	10.2	3.4	10.2	3.4

Food Waste	81	13.6	81	13.6
Notes: 1. PSD (Volatile Solids:76%, COD/VSS:1.58) 2. Domestic WAS (Volatile Solids:67%, COD/VSS:1.48) 3. Industrial WAS (Volatile Solids:62%, COD/VSS:1.56)				

(c) Primarily, Waste Activated Sludge (WAS) from DLM and ILM shall be fed to THP for digestion pre-treatment. In later stage of Phase 1, 30% of thickened primary sludge from DLM will be combined with the WAS for pre-dewatering and subsequently fed to THP process.

(d) These streams will be dewatered in Pre-dewatering centrifuges to achieve an average of around 17 percent total solids (TS) before feeding to the THP. Dewatered WAS (DWAS) cake will be transferred from the bottom of DWAS cake silo to THP for pre-treatment prior pumping to the Mesophilic Anaerobic Digester (MAD) for digestion. In the THP. the sludge will be heated to high temperature and pressure steam to disintegrate /dissolve the cell structures and organic materials to enhance the properties of sludge for MAD digestion and downstream dewatering.

P4D.2. THP FEED PARAMETERS

(a) The following summarises the envisaged design flows to THP Phase 1. The THP Phase 1 process plant shall be designed to treat the following capacity scenarios:

Table P4D- 2 Phase 1 Scenarios

Phase 1 Flows	Description
Scenario 1	WAS Only
Scenario 2	WAS + 30% PS from DLM

(b) As shown in Table P4D- 2, there will be two main scenarios to be considered in the design of the THP plant (Phase 1). Scenario 1 is envisaged during early stage where only WAS will be fed to THP plant, the maximum turndown is approximately 50% of the Scenario 2 design flows. The THP plant shall be designed to treat peak flows envisaged in Scenario 2 (Governing Case) Monthly Maximum (MM) case as shown in Table P4D- 3which shows the basis of feed to THP. The %DS from pre-dewatering of THP feed can be assumed as 16%ds at the feed to THP feed pumps but could vary between 16-18%ds during operation.

Table P4D- 3 THP Feed Flowrates

Parameter		Scenario 1: WAS Only		Scenario 2: WAS + Partial PSD	
		AA	MM	AA	MM (Governing Case)
WAS [1]	tDS/d	109.5 [1]	145.5 [1]	109.5[1]	145.4[1]
PSD [1]	tDS/d	0	0	41.2[1]	53.2[1]
Total [1]	tDS/d	109.5	145.5	150.7	198.6 [2]

Notes:

1. Nominal operating feed flowrates based on 90% of solids capture is considered for the performance of pre-dewatering centrifuge and additional 92% capture at PSD centrifuge upstream of pre-dewatering centrifuge.
2. This is the governing case. Peak capacity for Phase 1 THP plant shall be based on 98% solids capture in upstream pre-dewatering stages, i.e. 216 tDS/d. The resulting unit minimum design capacity for each THP module shall be 72 tDS/d.

(c) Table P4D- 4provides an overview of the Hydrolysed Sludge feed to the Digesters, during the MM and the %DS, the THP plant main benefit is to allow the MAD to operate within its design capacity and %DS. Hydrolysed sludge cooling is also required prior feeding to the MAD to maintain the Digesters' bulk temperature. The heat and material balance for overall biosolids plant are provided in Volume 5 Part-IA P&IDs for further information.

Table P4D- 4 Scenario 2 MM Overall feed to Digesters (For Information Only)

Feed stream	tDS/d	%DS
HS	198.6	13.6%
Thickened PSD	137.8	8.0%
Thickened PSI	61.8	8.0%
GW	10.2	3.4%
FW	81.0	13.5%
Combined Digester Feed	489.4	10.1%

END OF SECTION